https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	λ is the independent variable and h is the dependent variable **or** vary λ and measure h	**1**
	keep angle of incidence of light on to CD constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • light source, CD and screen • method to support light source • light source labelled and at least one other label	**1**
	method to change wavelength, e.g. use filters, different lasers/LEDs	**1**
	method to determine λ e.g. read from filter/label or experiment e.g. Young's slits, diffraction grating	**1**
	measure h with a rule **or** labelled rule shown in diagram close to screen with h indicated	**1**
	Method of analysis	
	plot a graph of h against λ **or** plot a graph of h against $n\lambda$	**1**
	$d = n$/gradient **or** $d = 1$/gradient (consistent with graph)	**1**
	$B = y$-intercept	**1**

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Question	Answer	Marks
	Additional detail including safety considerations	**Max. 6**
	D1 safety precaution linked to exposure of eyes to (intense) light, e.g. avoid looking (directly) at the (intense) light source or wear dark glasses or wear goggles to protect eyes from light	
	D2 keep distance from CD to screen constant	
	D3 keep *n* constant	
	D4 method to identify same order, e.g. identify the zero order and count the number of maxima to where *h* is measured	
	D5 equation to determine λ e.g. Young's slits/diffraction grating formula with λ as subject	
	D6 use intense light source/collimated light source	
	D7 perform experiment in a dark room	
	D8 clean surface of CD/check that there are no scratches	
	D9 detail on measuring *h* (e.g. measure to the top and bottom of the maximum and average)	
	D10 relationship valid if a straight line (with a *y*-intercept)	

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